AP Calculus BC Unit 9 Review Book

# AP Calculus BC Unit 9 Review Book

## Parametric Equations, Polar Coordinates, and Vector-Valued Functions

A complete English review guide for final AP review, written to be beginner-friendly, detailed, and exam-focused.

**Goal of this unit.** Unit 9 asks you to switch coordinate systems without losing the main calculus ideas. You will differentiate curves described by a parameter, describe motion with vectors, and compute slope and area in polar form. The most important habits are to use the correct variable, identify the correct geometric meaning, and sketch whenever the picture controls the setup.

| Chapter | Main idea |
|---|---|
| 9.1 | Parametric equations and tangent slopes |
| 9.2 | Second derivatives of parametric curves |
| 9.3 | Arc length in parametric form |
| 9.4-9.6 | Vector-valued functions and motion |
| 9.7 | Polar coordinates and slope in polar form |
| 9.8-9.9 | Polar area for one curve and two curves |

## Core Formula Sheet

Start with these formulas. The rest of the book explains how and when to use them.

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$$

$$\frac{d^2y}{dx^2} = \frac{\frac{d}{dt}\left(\frac{dy}{dx}\right)}{dx/dt}$$

$$L = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt$$

$$\vec{r}(t) = \langle x(t), y(t)\rangle, \quad \vec{v}(t) = \langle \dot{x}(t), \dot{y}(t)\rangle, \quad \vec{a}(t) = \langle \ddot{x}(t), \ddot{y}(t)\rangle$$

$$\text{speed} = |\vec{v}(t)| = \sqrt{(\dot{x}(t))^2 + (\dot{y}(t))^2}$$

$$x = r\cos\theta, \qquad y = r\sin\theta$$

$$\frac{dy}{dx} = \frac{\frac{dr}{d\theta}\sin\theta + r\cos\theta}{\frac{dr}{d\theta}\cos\theta - r\sin\theta}$$

$$A = \frac{1}{2}\int_\alpha^\beta r^2\, d\theta$$

$$A = \frac{1}{2}\int_\alpha^\beta \left(r_{\text{outer}}^2 - r_{\text{inner}}^2\right) d\theta$$

## 9.1 Defining and Differentiating Parametric Equations

A parametric curve is given by two equations, one for x and one for y, both in terms of the same parameter t. Think of t as the clock that moves the point along the curve. To find the slope of the curve, differentiate both coordinate functions with respect to t and then take their ratio.

$$x = f(t), \qquad y = g(t)$$

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$$

$$\text{horizontal tangent: } dy/dt = 0 \text{ and } dx/dt \neq 0$$
$$\text{vertical tangent: } dx/dt = 0 \text{ and } dy/dt \neq 0$$

- Always find dx/dt and dy/dt first. Most errors happen when students rush to the final ratio.
- A horizontal tangent needs a zero numerator and a nonzero denominator. A vertical tangent needs a zero denominator and a nonzero numerator.
- The same point can be traced at different parameter values, so always use the parameter value given in the question.

**Common AP mistakes.**

- Writing dy/dx = dy/dt and forgetting to divide by dx/dt.
- Finding dy/dt = 0 and forgetting to check that dx/dt is not zero.

**Worked example.** Find the slope and all horizontal tangents for x=t^2+1 and y=t^3-3t.

$$\frac{dx}{dt} = 2t, \qquad \frac{dy}{dt} = 3t^2 - 3$$

$$\frac{dy}{dx} = \frac{3t^2 - 3}{2t}$$

For horizontal tangents, set the numerator equal to zero.

$$3t^2 - 3 = 0 \Rightarrow t = \pm 1$$

Now check the denominator. Since 2(1) and 2(-1) are both nonzero, both values work. The corresponding points are (2,-2) and (2,2).

## 9.2 Second Derivatives of Parametric Equations

For a parametric curve, the second derivative is found in two layers. First compute dy/dx as a function of t. Then differentiate that result with respect to t, and divide by dx/dt again. This tells you the concavity of the curve with respect to x.

$$\frac{d^2y}{dx^2} = \frac{\frac{d}{dt}\left(\frac{dy}{dx}\right)}{dx/dt}$$

- This formula comes from the chain rule. It is not the same as dividing second derivatives with respect to t.
- Use the sign of d squared y / d x squared to decide whether the curve is concave up or concave down at a parameter value.

**Common AP mistakes.**

• Using (d squared y / d t squared) divided by (d squared x / d t squared). That is not the correct formula.

• Forgetting to simplify dy/dx before differentiating it with respect to t.

**Worked example.** Find d squared y / d x squared for x=t^2 and y=t^3.

$$\frac{dx}{dt} = 2t, \qquad \frac{dy}{dt} = 3t^2$$

$$\frac{dy}{dx} = \frac{3t^2}{2t} = \frac{3}{2}t$$

$$\frac{d}{dt}\left(\frac{dy}{dx}\right) = \frac{3}{2}$$

$$\frac{d^2y}{dx^2} = \frac{\frac{3}{2}}{2t} = \frac{3}{4t}$$

## 9.3 Finding Arc Lengths of Curves Given by Parametric Equations

Arc length for a parametric curve is the integral of the speed of the moving point. Once you know how fast x and y change, you combine them with the Pythagorean Theorem.

$$L = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt$$

- The limits must be parameter values.
- The integrand is the same expression used for speed in motion problems.
- If the expression under the square root simplifies to 1, the problem becomes especially clean.

**Common AP mistakes.**

- Using dx or dy instead of dt in the integral.
- Forgetting to square both derivatives before adding them.

**Worked example.** Find the arc length of x=\cos t and y=\sin t on 0\le t\le \pi.

$$\frac{dx}{dt} = -\sin t, \qquad \frac{dy}{dt} = \cos t$$

$$L = \int_0^{\pi} \sqrt{\sin^2 t + \cos^2 t}\, dt$$

$$L = \int_0^{\pi} 1\, dt = \pi$$

This matches the upper semicircle of radius 1, whose length is also pi.

## 9.4 Defining and Differentiating Vector-Valued Functions

A vector-valued function stores several coordinate functions in one object. In AP Calculus BC, this is usually a position vector. Differentiate a vector by differentiating each component separately.

$$\vec{r}(t) = \langle x(t), y(t) \rangle \quad \text{or} \quad \vec{r}(t) = \langle x(t), y(t), z(t) \rangle$$

$$\vec{v}(t) = \langle \dot{x}(t), \dot{y}(t) \rangle$$

- If r(t) is position, then its derivative is velocity.
- Keep the angle brackets. They remind you that the result is still a vector, not a scalar.

**Common AP mistakes.**

- Dropping vector notation and treating the whole expression as a single ordinary function.
- Mixing up velocity with speed. Velocity is a vector; speed is a magnitude.

**Worked example.** Differentiate r(t)=.

$$\vec{v}(t) = \langle 2t, e^t, \cos t \rangle$$

## 9.5 Integrating Vector-Valued Functions

Integrating a vector-valued function is also component-by-component. Include a constant vector, then use any given initial condition to determine its components.

$$\int \vec{r}(t)\,dt = \left\langle \int x(t)\,dt, \int y(t)\,dt \right\rangle + \vec{C}$$

- Use the whole initial vector when solving for constants.
- This is the standard method for recovering velocity from acceleration or position from velocity.

**Common AP mistakes.**

- Forgetting the constant vector.
- Using only one constant when the vector has two or three components.

**Worked example.** Find r(t) if v(t)=<2t,\cos t> and r(0)=<1,2>.

$$\vec{r}(t) = \langle t^2, \sin t\rangle + \langle C_1, C_2\rangle$$

$$\vec{r}(0) = \langle 0, 0\rangle + \langle C_1, C_2\rangle = \langle 1, 2\rangle$$

$$\vec{r}(t) = \langle t^2 + 1, \sin t + 2\rangle$$

## 9.6 Solving Motion Problems Using Parametric and Vector-Valued Functions

Motion in the plane uses the same hierarchy as one-dimensional motion. Position differentiates to velocity, velocity differentiates to acceleration, and the magnitude of the velocity vector gives speed.

$$\vec{v}(t) = \langle \dot{x}(t), \dot{y}(t)\rangle, \qquad \vec{a}(t) = \langle \ddot{x}(t), \ddot{y}(t)\rangle$$

$$\text{speed} = \sqrt{(\dot{x}(t))^2 + (\dot{y}(t))^2}$$

- Speed is always nonnegative because it is a magnitude.
- Distance traveled uses the speed formula; displacement uses the change in position vector or the coordinate changes directly.

**Common AP mistakes.**

- Calling the vector velocity the speed.
- Forgetting the square root when finding speed.

**Worked example.** For r(t)=, find velocity, acceleration, and speed.

$$\vec{v}(t) = \langle 2t, 3t^2 \rangle$$

$$\vec{a}(t) = \langle 2, 6t \rangle$$

$$\text{speed} = \sqrt{(2t)^2 + (3t^2)^2} = \sqrt{4t^2 + 9t^4}$$

## 9.7 Defining Polar Coordinates and Differentiating in Polar Form

Polar coordinates describe a point by a distance from the origin and an angle from the positive x-axis. To differentiate in polar form, convert to x and y using theta as the parameter, then use the standard parametric slope formula.

$$x = r\cos\theta, \qquad y = r\sin\theta$$

$$\frac{dx}{d\theta} = \frac{dr}{d\theta}\cos\theta - r\sin\theta$$

$$\frac{dy}{d\theta} = \frac{dr}{d\theta}\sin\theta + r\cos\theta$$

$$\frac{dy}{dx} = \frac{\frac{dr}{d\theta}\sin\theta + r\cos\theta}{\frac{dr}{d\theta}\cos\theta - r\sin\theta}$$

• If you forget the formula, rebuild it from x=r cos theta and y=r sin theta. That is the safest method on an exam.

- Horizontal tangents require dy/d theta = 0 and dx/d theta not equal to zero. Vertical tangents require the opposite.
- The same point can have multiple polar descriptions, so a quick sketch is often essential.

**Common AP mistakes.**

- Trying to differentiate r with respect to x directly.
- Setting only one of the tangent conditions and forgetting to check the other.

**Worked example.** Find the slope for the polar curve r=2 heta.

$$\frac{dr}{d\theta} = 2$$

$$\frac{dy}{dx} = \frac{2\sin\theta + 2\theta\cos\theta}{2\cos\theta - 2\theta\sin\theta}$$

$$\frac{dy}{dx} = \frac{\sin\theta + \theta\cos\theta}{\cos\theta - \theta\sin\theta}$$

## 9.8 Finding the Area of a Polar Region or the Area Bounded by a Single Polar Curve

A polar area is built from thin sectors, not thin rectangles. That is why the area formula includes one-half times r squared. The hardest part is usually choosing the correct theta-interval.

$$A = \frac{1}{2}\int_{\alpha}^{\beta} r^2 \, d\theta$$

- To find one petal or one loop, solve r=0 and use the relevant pair of angles that trace exactly that part of the graph.
- Sketching is strongly recommended. A good sketch usually prevents wrong limits.

**Common AP mistakes.**

- Forgetting the factor one-half.
- Integrating over a full cycle when the question asks for only one petal or one loop.

**Worked example.** Set up the area of one petal of r=\sin(2 heta).

Start where the petal begins and ends, which is where r equals zero.

$\sin(2\theta) = 0 \Rightarrow \theta = 0$ and $\theta = \frac{\pi}{2}$ for one positive petal interval

$$A = \frac{1}{2}\int_{0}^{\pi/2} \sin^2(2\theta) \, d\theta$$

## 9.9 Finding the Area of the Region Bounded by Two Polar Curves

Area between two polar curves behaves like a washer problem in theta: outer radius squared minus inner radius squared. The real work is deciding which curve is farther from the origin on each interval and whether the interval must be split.

$$A = \frac{1}{2}\int_{\alpha}^{\beta} \left(r_{\text{outer}}^2 - r_{\text{inner}}^2\right) d\theta$$

- Find intersection angles first, but do not stop there. You still need to decide which curve is outer on the interval you plan to use.

- Many AP problems require a sketch because the outer curve can switch.

**Common AP mistakes.**

- Assuming one curve stays outer across the whole interval without checking.
- Forgetting that the origin can also serve as a boundary point.

**Worked example.** Set up the comparison for the curves r=2 and r=2\sin heta.

First solve the intersection equation.

$$2 = 2\sin\theta \Rightarrow \sin\theta = 1 \Rightarrow \theta = \frac{\pi}{2}$$

Now sketch both curves. The correct area setup depends on which specific region is requested, because the outer and inner radii can change with theta. The picture decides the correct interval and subtraction order.

**Unit 9 quick checklist before a test.** For parametric problems, compute x dot and y dot first, then build slope, tangent conditions, speed, and arc length from those two derivatives.

For vector problems, differentiate or integrate each component separately and interpret the result carefully as position, velocity, acceleration, or speed.

For polar problems, sketch the curve, use x=r cos theta and y=r sin theta when needed, and remember that area formulas use one-half.